

10.4 Practice Questions.

17 $\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n} + \sqrt[3]{n}}$ $\sqrt[n]{a} = (a)^{1/n}$

Sol

$\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n} + \sqrt[3]{n}} = \sum_{n=1}^{\infty} a_n$ either you can apply basic comparison test or

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} b_n$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{\frac{1}{2\sqrt{n} + \sqrt[3]{n}}}{\frac{1}{\sqrt{n}}} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{2\sqrt{n} + \sqrt[3]{n}} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}/\sqrt{n}}{2\sqrt{n}/\sqrt{n} + \frac{\sqrt[3]{n}}{\sqrt{n}}} \\ &= \lim_{n \rightarrow \infty} \frac{1}{2 + \frac{\sqrt[3]{n}}{\sqrt{n}}} \end{aligned}$$

$$\therefore \frac{\sqrt[3]{n}}{\sqrt{n}} = \frac{n^{1/3}}{n^{1/2}} = \frac{1}{n^{1/2 - 1/3}} = \frac{1}{n^{1/6}}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2 - \frac{1}{n^{1/6}}} = \frac{1}{2} > 0$$

So by P-S-T

$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \cdot n = \frac{1}{2}$ diverges

So by L.C.T $\sum_{n=1}^{\infty} a_n$ diverges

18 $\sum_{n=1}^{\infty} \frac{3}{n + \sqrt{n}} = \sum_{n=1}^{\infty} a_n$

Sol

$\sum_{n=1}^{\infty} \frac{1}{n} = \sum b_n$

$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{3}{n + \sqrt{n}}}{\frac{1}{n}}$

D: Y.S.

Q18 $\sum_{n=1}^{\infty} \frac{\sin^2 n}{2^n}$

Solution:-

$$\sum_{n=1}^{\infty} \frac{\sin^2 n}{2^n}$$

$$-1 \leq \sin n \leq 1$$

$$0 \leq \sin^2 n \leq 1$$

$$\frac{0}{2^n} \leq \frac{\sin^2 n}{2^n} \leq \frac{1}{2^n}$$

So $\sum_{n=1}^{\infty} \frac{1}{2^n} = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$; $|r| < 1$
by Geometric series test $r = \frac{1}{2} < 1$

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \text{ converges}$$

So $\sum_{n=1}^{\infty} \frac{\sin^2 n}{2^n} \text{ converges}$

Q20

$$\sum_{n=1}^{\infty} \frac{1 + \cos n}{n^2}$$

Solution

$$-1 \leq \cos n \leq 1$$

$$1-1 \leq 1 + \cos n \leq 1+1$$

$$0 \leq \frac{1 + \cos n}{n^2} \leq \frac{2}{n^2}$$

D.Y.S

Q21

$$\sum_{n=1}^{\infty} \frac{2^n}{3n-1}$$

Solution.

Powers are same so n^{th} term test is valid

$$\sum_{n=1}^{\infty} \frac{2^n}{3n-1} = \sum_{n=1}^{\infty} a_n$$

$$a_n = \frac{2^n}{3n-1}$$

$$\lim_{n \rightarrow \infty} \frac{2}{3} = \frac{2}{3} \neq 0 \text{ so by divergent test it is divergent.}$$

$$Q22 \quad \sum_{n=1}^{\infty} \frac{n+1}{n^2 \sqrt{n}}$$

Solution

$$\sum_{n=1}^{\infty} \left(\frac{1}{n^{2(n^{1/2})}} + \frac{1}{n^2 \sqrt{n}} \right)$$

$$\sum_{n=1}^{\infty} \left(\frac{1}{n^{3/2}} + \frac{1}{n^{5/2}} \right)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}} + \sum_{n=1}^{\infty} \frac{1}{n^{5/2}}$$

for $p=3/2$

$$p=5/2$$

Series converges

2nd Method

$$b_n = \frac{n}{n^{5/2}} = \frac{1}{n^{3/2}}$$

Q23

$$\sum_{n=1}^{\infty} \frac{10n+1}{n(n+1)(n+2)}$$

Solution: $\sum_{n=1}^{\infty} \frac{10n+1}{n(n^2+2n+n+2)}$

$$\sum_{n=1}^{\infty} \frac{10n+1}{n(n^2+3n+2)}$$

$$\sum_{n=1}^{\infty} \frac{10n+1}{n^3+3n^2+2n} = \sum_{n=1}^{\infty} a_n$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} b_n$$

So: $\lim_{n \rightarrow \infty} a_n/b_n$

$$= \lim_{n \rightarrow \infty} \frac{2n+1}{n^3+3n^2+2n}$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{1}{n^2}}{\frac{2n^3+n^2}{n^3+3n^2+2n}} = \lim_{n \rightarrow \infty} \frac{2n^3/n^3 + n^2/n^3}{n^3/n^3 + 3n^2/n^3 + 2n/n^3}$$

$$= \lim_{n \rightarrow \infty} \frac{2 + 1/n}{1 + 3/n + 2/n^2} = \frac{2 + 1/\infty}{1 + 3/\infty + 2/\infty} = \frac{2}{1} > 1$$

So by P-S-T

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^p} \Rightarrow p=2 > 1 \text{ convergent}$$

So by L.C.T $\sum_{n=1}^{\infty} \frac{10n+1}{n(n+1)(n+2)}$

Example:-

$$\sum_{n=3}^{\infty} \frac{5n^3-3n}{n^2(n-2)(n^2+5)} =$$

Solution $\sum_{n=3}^{\infty} \frac{5n^3-3n}{n^2(n^3+5n-2n^2-10)} = \sum_{n=1}^{\infty} \frac{5n^3-3n}{n^5+5n^3-2n^4-10n^2} = \sum_{n=1}^{\infty} \frac{5n^3-3n}{n^5+5n^3-2n^4-10n^2} = \sum a_n$

$$\sum_{n=1}^{\infty} \frac{5n^3}{n^5} = \sum_{n=1}^{\infty} \frac{5}{n^2} = \sum b_n$$

DY.S

Q 25

$$\sum_{n=1}^{\infty} \left(\frac{n}{3n+1} \right)^n$$

Solution:-

$$\sum_{n=1}^{\infty} \left(\frac{n}{3n+1} \right)^n = \sum_{n=1}^{\infty} a_n$$

$$\sum_{n=1}^{\infty} \left(\frac{n}{3n} \right)^n = \sum_{n=1}^{\infty} \left(\frac{1}{3} \right)^n = \sum b_n$$

$$\text{So } \lim_{n \rightarrow \infty} \frac{\left(\frac{n}{3n+1} \right)^n}{\left(\frac{1}{3} \right)^n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n}{3n+1} \right)^n 3^n$$

$$= \lim_{n \rightarrow \infty} \left(\frac{3n}{3n+1} \right)^n$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} e^{n \ln \left(\frac{3n}{3n+1} \right)}$$

$$= e \lim_{n \rightarrow \infty} n \ln \left(\frac{3n}{3n+1} \right)$$

$$= e \lim_{n \rightarrow \infty} \frac{\ln \left(\frac{3n}{3n+1} \right)}{\frac{1}{n}}$$

2nd Method

$$\sum_{n=1}^{\infty} \left(\frac{n}{3n+1} \right)^n$$

$$3n = 3n$$

$$3n+1 > 3n$$

$$\frac{1}{3n+1} < \frac{1}{3n}$$

$$\frac{n}{3n+1} < \frac{n}{3n}$$

$$\left(\frac{n}{3n+1} \right)^n < \frac{1}{3^n}$$

$a_n < b_n \rightarrow$ by geometric series

Q 26

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+2}} = \sum_{n=1}^{\infty} a_n$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}} = \sum_{n=1}^{\infty} b_n$$

Q26 $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+2}}$

Sol

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+2}} = \sum a_n$$

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3}} = \sum b_n$$

$$n^3 = n^3$$

$$n^3 + 2 > n^3$$

$$\frac{1}{n^3+2} < \frac{1}{n^3}$$

$$\frac{1}{\sqrt{n^3+2}} < \frac{1}{\sqrt{n^3}} = \frac{1}{n^{3/2}}$$

By p-series test

$$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}} = \sum_{n=1}^{\infty} \frac{1}{n^p}$$

$p = 3/2 > 1$ convergent.

So by direct comparison test

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+2}} \text{ is } \text{gt.}$$

Q 27

$$\sum_{n=1}^{\infty} \frac{1}{\ln(\ln n)}$$

Hint $n > \ln n > \ln(\ln n)$

Solution:-

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

$$n > \ln n > \ln(\ln n)$$

$$n > \ln(\ln n)$$

$$\frac{1}{n} < \frac{1}{\ln(\ln n)}$$

So by P-S-T

$$\sum_{n=1}^{\infty} \frac{1}{n} = \sum_{n=1}^{\infty} b_n$$

$p=1$ divergent

So by Comparison test

$$\sum_{n=1}^{\infty} a_n \text{ converges}$$

Q 28

$$\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^3}$$

$$\frac{(\ln n)^2}{n^3} \bigg/ \frac{1}{n^3} = (\ln n)^2 = \infty$$

$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ if b_n diverges
but here b_n converges

So we cannot take $b_n = \frac{1}{n^3}$

Let's try with $\frac{1}{n^2}$

Sol

$$\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^3}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{(\ln n)^2/n^3}{1/n^2} = \lim_{n \rightarrow \infty} \frac{(\ln n)^2}{n} \left(\frac{\infty}{\infty} \right) \text{ Apply DL Hopital Rule}$$

$$\lim_{n \rightarrow \infty} \frac{2 \ln n \times \frac{1}{n}}{1} = \lim_{n \rightarrow \infty} \frac{\ln n}{n} \left(\frac{\infty}{\infty} \right) = \lim_{n \rightarrow \infty} \frac{1/n}{1} = \frac{1}{\infty} = 0 \Rightarrow \text{So } \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges so by L.C.T. } \sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^3} \text{ converges}$$

Q29

$$\sum_{n=2}^{\infty} \frac{1}{\ln \ln n}$$

$$b_n = \frac{1}{\ln n} \quad \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{1}{\ln n} = \frac{1}{\infty} = 0 \quad \text{as } b_n \text{ diverges} \Rightarrow \text{no conclusion about } a_n.$$

Solⁿ Let

$$\sum_{n=2}^{\infty} \frac{1}{\ln \ln n} = \sum_{n=2}^{\infty} a_n$$

$$\sum_{n=2}^{\infty} \frac{1}{n} = \sum_{n=2}^{\infty} b_n$$

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{\ln \ln n}}{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{n}{\ln \ln n} = \lim_{n \rightarrow \infty} \frac{\ln n}{\ln n} \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{n}} = \lim_{n \rightarrow \infty} n = \infty$$

So limit is infinity.

by P-S-T b_n diverges

So by L-S-T a_n also diverges

Q30

$$\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^{3/2}}$$

Solution:-

$$\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^{3/2}} = \sum_{n=1}^{\infty} a_n$$

$$= \sum_{n=1}^{\infty} \frac{1}{n^{5/4}} = \sum_{n=1}^{\infty} b_n$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{(\ln n)^2}{n^{3/2}} \div \frac{1}{n^{5/4}} = \lim_{n \rightarrow \infty} \frac{n^{5/4} (\ln n)^2}{n^{3/2}} = \lim_{n \rightarrow \infty} \frac{(\ln n)^2}{n^{1/4}} \frac{\infty}{\infty}$$

DYS

Q31

$$\sum_{n=1}^{\infty} \frac{1}{1 + \ln n}$$

Solution:-

$$\sum_{n=1}^{\infty} \frac{1}{1 + \ln n} = \sum a_n$$

$$\sum_{n=1}^{\infty} \frac{1}{n} = \sum b_n$$

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{1 + \ln n}}{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{n}{1 + \ln n}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{1/n} = n = \infty$$

So $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges

So by L.C.T $\sum_{n=1}^{\infty} a_n$ ~~converges~~ ^{diverges}

Q32

$$\sum_{n=2}^{\infty} \frac{\ln(n+1)}{n+1}$$

P, C, D $\frac{\ln(n+1)}{n+1}$ Positive for $n \geq 1$; it is continuous at $[1, \infty)$
By taking derivative $\frac{\ln(n+1)}{n+1}$ decreasing in

$$\int_2^{\infty} \frac{\ln(x+1)}{1+x} dx = \left. \frac{(\ln(x+1))^2}{2} \right|_2^{\infty}$$

$$= \frac{1}{2} [\ln \infty - \ln 3] = \infty \text{ so by integral test}$$

$$\sum_{n=2}^{\infty} \frac{\ln(n+1)}{n+1} \text{ diverges.}$$

$$\left\{ \begin{aligned} f(x) &= \frac{\ln(x+1)}{x+1} \\ f'(x) &= \frac{1 - \ln(x+1)}{(1+x)^2} \leq 0 \end{aligned} \right.$$

$$1 - \ln(x+1) \leq 0$$

$$1 \leq \ln(x+1)$$

$$e = e^{\ln(x+1)}$$

$$e = x+1$$

$$e-1 = x$$

$$1.71 = x$$

$$Q = \sum_{n=2}^{\infty} \frac{1}{n\sqrt{n^2-1}}$$

Q

$$n^2 > 17n \quad ; \quad n > 17$$

$$n^2(n^2-1) > n^3$$

$$\sqrt{n^2(n^2-1)} > \sqrt{n^3}$$

$$\frac{1}{\sqrt{n^2(n^2-1)}} < \frac{1}{n^{3/2}}$$

$$\frac{1}{n\sqrt{n^2-1}} \leq \frac{1}{n^{3/2}}$$

So by P.S.T $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges

So by L.C.T $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n^2-1}}$ also converges.

Q

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$$

Solution:-

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1} = \sum a_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$$

apply L.C.T

$$\sum_{n=1}^{\infty} \frac{a_n}{b_n} = \sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1} \div \frac{1}{n^{3/2}}$$

$$= \sum_{n=1}^{\infty} \frac{n^2}{n^2+1} = \lim_{n \rightarrow \infty} \frac{n^2/n^2}{n^2/n^2 + 1/n^2} = \lim_{n \rightarrow \infty} \frac{1}{1 + 1/n^2} = 1 \neq 0$$

as by P.S.T $\sum_{n=1}^{\infty} b_n$ converges by P.S.T

So by L.C.T $\sum_{n=1}^{\infty} a_n$ converges.

Q 35

$$\lim_{n \rightarrow \infty} \frac{1-n}{n2^n} = \lim_n$$

Solution:-

$$\sum_{n=1}^{\infty} \frac{1}{n2^n} = \sum_{n=1}^{\infty} \frac{n'}{n2^n}$$

by geometric series
 $\sum_{n=1}^{\infty} \frac{1}{2^n}$ converges.

$$2^n = 2^n$$

$$n2^n \geq 2^n$$

$$\frac{1}{n2^n} \leq \frac{1}{2^n}$$

Q 36 same case like Q 35

$$\sum_{n=1}^{\infty} \frac{n+2^n}{n^2 2^n}$$

$$\sum_{n=1}^{\infty} \left(\frac{n}{n^2 2^n} + \frac{2^n}{n^2 2^n} \right)$$

DTS

Q 37

$$\frac{1}{3^{n-1} + 1}$$

Q 38

$$\sum_{n=1}^{\infty} \frac{3^{n-1} + 1}{3^n}$$

$$\sum_{n=1}^{\infty} \frac{3^{n-1}}{3^n} \rightarrow \frac{1}{3}$$

Q 39

$$\sum_{n=1}^{\infty} \frac{n+1}{n^2+3n} \cdot \frac{1}{5^n}$$

Sol

$$\sum_{n=1}^{\infty} \frac{n+1}{n^2+3n} \cdot \frac{1}{5^n} = \lim_{n \rightarrow \infty} \frac{n+1}{n^2+3n} = \lim_{n \rightarrow \infty} \frac{\frac{n}{n^2} + \frac{1}{n^2}}{\frac{n^2}{n^2} + \frac{3n}{n^2}} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n} + \frac{1}{n^2}}{1 + \frac{3}{n}} = 0$$

Hence by comparison test $\sum_{n=1}^{\infty} b_n$ converges as $\sum_{n=1}^{\infty} a_n$ is.

Q40

$$\sum_{n=1}^{\infty} \frac{2^n + 3^n}{3^n + 4^n}$$

Solution:-

$$\sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n \Rightarrow \sum_{n=1}^{\infty} \frac{3^n}{4^n} = \sum b_n$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{2^n + 3^n}{3^n + 4^n} \times \frac{4^n}{3^n}$$

$$= \lim_{n \rightarrow \infty} \frac{(8/12)^n + 1}{(9/12)^n + 1}$$

$$= \frac{0+1}{0+1} \quad \text{as } \lim_{n \rightarrow \infty} n^n \approx |n| \leq 1 \text{ otherwise } \infty.$$

= 170

So by Geometric series test $(\frac{3}{4})^n$ converges

So by L.C.T $\sum a_n$ converges.

Q41

$$\sum_{n=1}^{\infty} \frac{2^n - n}{n2^n}$$

Sol

$$\sum_{n=1}^{\infty} \left(\frac{2^n}{n2^n} - \frac{n}{n2^n} \right)$$

$$\sum_{n=1}^{\infty} \frac{1}{n} - \sum_{n=1}^{\infty} \frac{1}{2^n}$$

div - con = div

Q42

$$\sum_{n=1}^{\infty} \frac{\ln n}{n e^n} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{e^n}$$

$$\text{Sol} \quad \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\ln n}{n e^n} \times e^n = \lim_{n \rightarrow \infty} \frac{\ln n}{n} \frac{\infty}{\infty}$$

$$= \lim_{n \rightarrow \infty} \frac{1/n}{1/2n} = \lim_{n \rightarrow \infty} \frac{1}{2n} = 0$$

So if $\sum b_n$ converges then $\sum a_n$ also

$$Q \quad \sum_{n=2}^{\infty} \frac{1}{n!}$$

$$n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$$

$$n! \geq n(n-1)$$

$$\frac{1}{n!} \leq \frac{1}{n(n-1)}$$

$$\sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} b_n \quad \text{This is telescopic}$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n-1} - \frac{1}{n} \right)$$

$$S_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right)$$

$$= 1 - \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1 \text{ converges}$$

If $\sum b_n$ converges then $\sum a_n$ also

Q44

$$\sum_{n=1}^{\infty} \frac{(n-1)!}{(n+2)!}$$

$$\underline{Ex} \quad \frac{(n-1)!}{(n+2)!}$$

$$\underline{Sol} \quad \sum_{n=1}^{\infty} \frac{(n-1)!}{(n+2)(n+1)(n)(n-1)!}$$

$$\sum_{n=1}^{\infty} \frac{1}{(n+2)(n+1)n}$$

$$\underline{Example:-} \quad \sum_{n=1}^{\infty} \sin \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n}}{\frac{1}{n}} = 0$$

$$\sum_{n=1}^{\infty} \sin \frac{1}{n} = \sum a_n$$

$$\sum_{n=1}^{\infty} \frac{1}{n} = \sum b_n$$

$$\text{also } \lim_{n \rightarrow \infty} \frac{a_n}{b_n} =$$

$$\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n}}{\frac{1}{n}} = 1 > 0$$

So by harmonic series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ div

and by L.C.T $\sum_{n=1}^{\infty} a_n$ div

$$\underline{Ex} \quad \sum_{n=1}^{\infty} \tan \frac{1}{n}$$

$$\underline{Ex} \quad \sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^{1.1}}$$

$$\tan^{-1} n < \frac{\pi}{2}$$

$$\frac{\tan^{-1} n}{n^{1.1}} < \frac{\frac{\pi}{2}}{n^{1.1}}$$

Example 48

$$\sum_{n=1}^{\infty} \frac{\sec n}{n^{1.3}}$$

Q49 $\sum \frac{\cosh n}{n^2} = \sum a_n$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} b_n$$

$$\lim_{n \rightarrow \infty} \frac{\cosh n}{n^2} \Big/ \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

$$\lim_{n \rightarrow \infty} \frac{e^{2n} + 1}{e^{2n} - 1} = \frac{e^{2n}(1 + e^{-2n})}{e^{2n}(1 - e^{-2n})} = 170$$

Q53 $\sum_{n=1}^{\infty} \frac{1}{1+2+3+\dots+n} = \sum_{n=1}^{\infty} \frac{2}{n(n+1)}$

Q50 $\frac{\tanh n}{n^2}$ DYS

Q51 $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{\sqrt{n}}}}$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n}} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{1} = 170$$

So if ~~$\sum_{n=1}^{\infty} \frac{1}{n}$~~ diverges so $\sum a_n$ also

Q54 $\sum_{n=1}^{\infty} \frac{1}{1+2^2+3^2+4^2+\dots+n^2}$

$$\sum_{n=1}^{\infty} \frac{6}{n(n+1)(2n+1)}$$

Q52 $\sum_{n=1}^{\infty} \frac{\sqrt[n]{n}}{n^2} = \sum a_n$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = b_n$$